ANA 322 Meninges

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Cerebral meninges: Connective tissue wrapping CNS intervening b/w it and surrounding bone. Together with fluid that they enclose cushion the nervous system.

They are three layers each from brain to spinal cord; (i) outer dura mater (pachymeninx), (ii) Middle arachnoid mater, (iii) Inner pia mater.

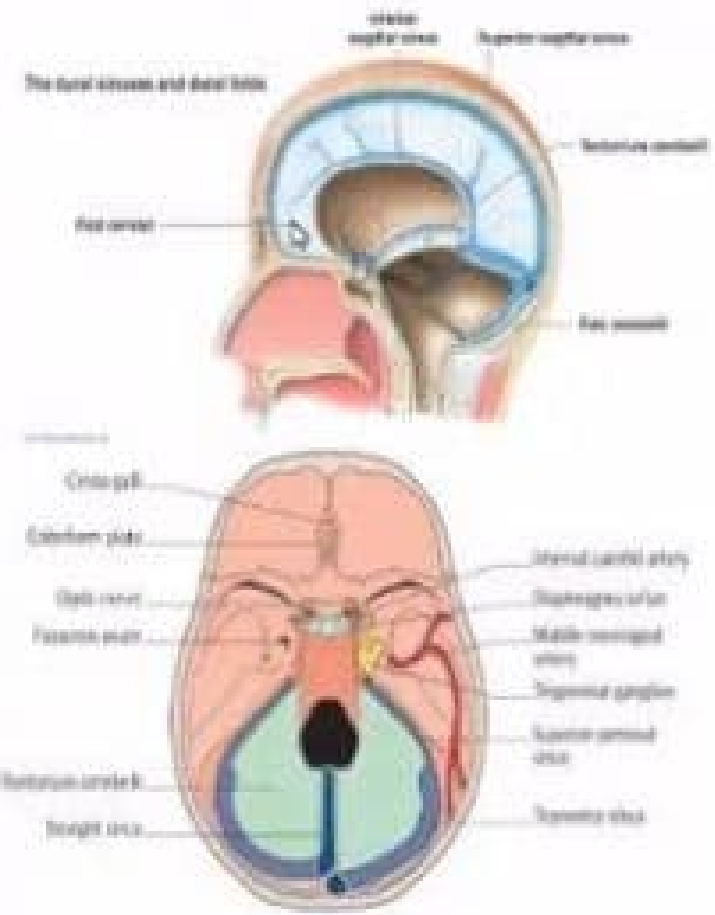
Arachnoid and pia mater together are called lepto-meninges.

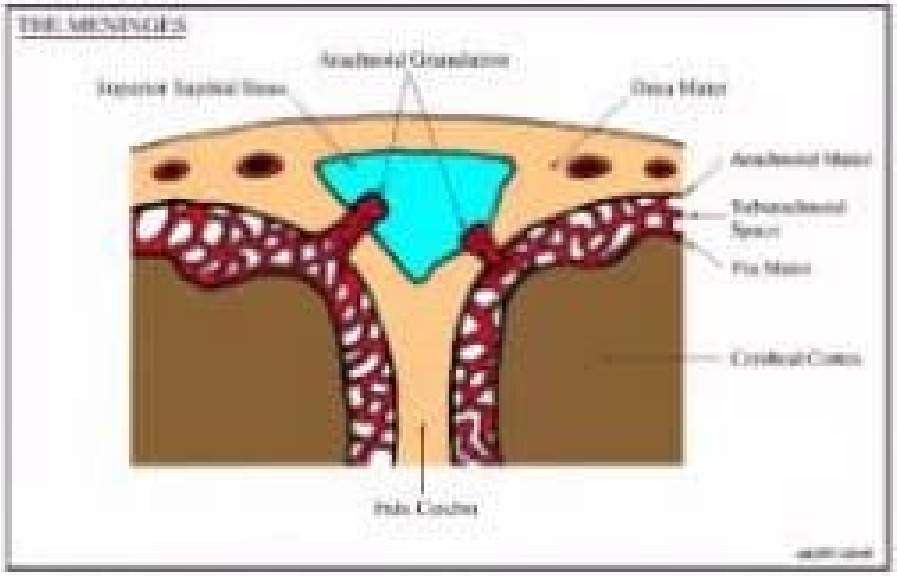
Cerebral Dural mater: Outermost thickest and toughest membrane covering the brain.

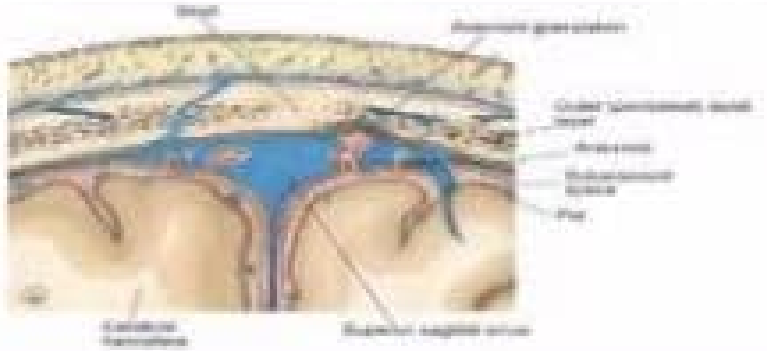
Has two layers, outer or endosteal layer and inner meningeal layer.

Endosteum or endocranium: (a) Continues with periosteum lining the outer side of the skull through sutures or foramens; (b) Periosteum lining the orbit through the superior orbital fissure; (C) Provides sheath for cranial nerve; (d) Its outer surface adherent to the inner surface of the cranial bones by a number of fine fibrous and vascular process with adherence most marked at the sutures.

Inner meningeal layer: Surrounds the brain and continues with spinal dural mater. The two layers fused except where they enclose venous sinus. It is folded to its self to form partitions that divides the cranial cavity into compartments. Folds include falx cerebri, tetorium cerebelli and falx cerebelli.


Internal carotid artery
Superior sagittal sinus
The dural sinuses and dural folds
Tentorium cerebelli
Pituitary gland
Falx cerebri
Crista galli
Cerebellum (cut)
Optic nerve
Pituitary gland
Internal carotid artery
Diaphragma sellae
Middle meningeal artery
Trigeminal ganglion
Superior petrosal sinus
Transverse sinus
Tentorium cerebelli
Straight sinus


THE MENINGES
Arachnoid Granulations
Superior Sagittal Sinus
Dura Mater
Arachnoid Mater
Subarachnoid Space
Pia Mater
Cerebral Cortex
Falx Cerebri


Skull
Pia
Superior sagittal sinus

Falx cerebri: Layer sickle shaped of dura mater occupying the median longitudinal b/w two central hemisphere. Has two ends; anterior end narrowed attached to crista galli. Posterior end broad and attached along medial plain to upper surface tentorium cerebelli.

Falx Cerebelli: Small sickle-shaped fold of dura projecting forward into posterior cerebellar notch. Base of sickle attached to the posterior part of inferior surface of tentorium cerebelli in the median plane. The apex is frequently divided into two parts that usually disappears at the sides of foramen magnum. The posterior margin is convex attached to inner occipital crest enclosing occipital sinus. Anterior margin concave and free.

Diaphragma sellae: Small circular horizontal fold of dura mater that forms the roof of hypophyseal fossa. Anteriorly attached to tuberculum sellae and posteriorly to dorsum sellae.

Tentorium cerebelli: Stretches horizontally like a roof over posterior cranial fossa and separates the cerebellum from the underlying cerebrum while vertical falx cerebelli partially separates the right and left halves of cerebellum at the inferior surface.

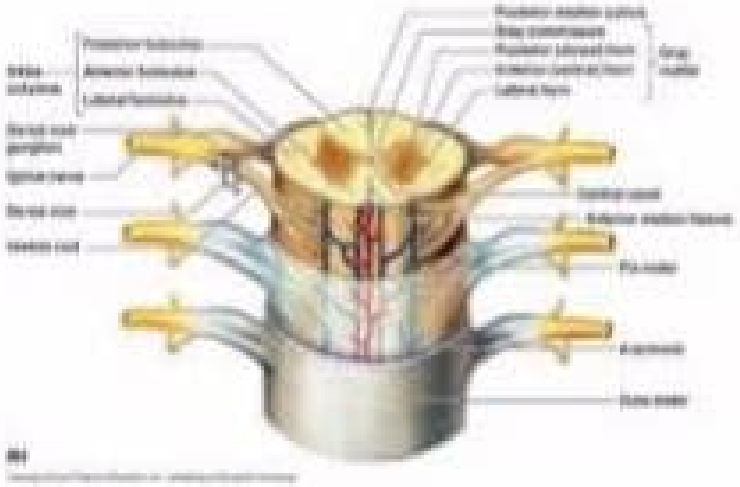
Arachnoid mater: A transparent membrane loosely surround the brain without dipping into the sulci. It bridges all irregularities of brain except the stem of lateral sulcus. It can not be identified in the hypophyseal fossae.

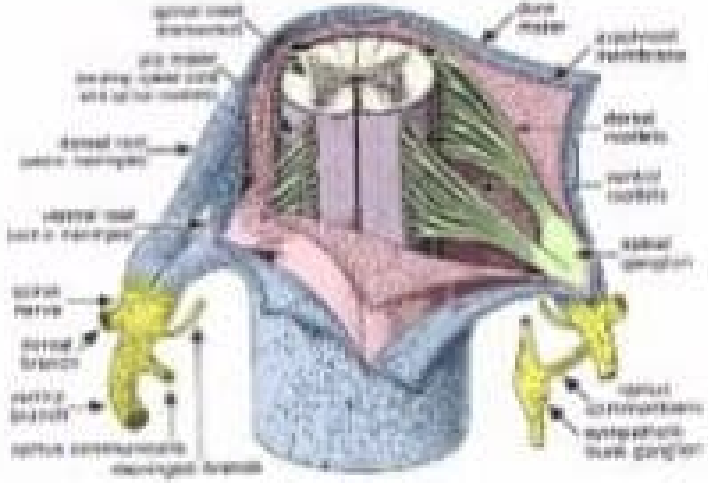
It is separated from dura by sub-dural space and from pia from sub-arachnoid space.

- Provides sheath for cranial nerve as far as their exit from skull.
- Has small finger-like projection, villi into cranial venous sinuses which absorb CSF.
- Villi enlarges in size with age to form arachnoid granulations which might produce depression in the inner skull.

Pia mater: Thin vascular membrane closely investing the brain dipping into various sulci and other irregularities of its surface.

- Provides perivascular sheath for minute vessels entering or leaving the brain substance.
- Its fold encloses the turf of capillaries forming tela choroidea.





Spinal meninges: It is made up of three sheaths of fibrous tissue that surrounds the spinal cord. The outer dura, middle arachnoid and inner pia maters.

Dura mater: A prolongation of meningeal layer of dura mater of posterior cranial fossa. The sheath extend down through foramen magnum to which it is attached and then extends to the level of S2 vertebra.

- It is separated from spinal canal by a layer of fat in which lies the internal vertebral venous plexus. It extends downward as sheath larger than the cord.
- It narrows downward fusing with filum terminalis at 2nd sacral segment.

- It is pierced segmentally by anterior and posterior roots of spinal nerve making prolongation over the nerve root.
- Between it and arachnoid is subdural space.

Arachnoid mater: A middle covering, thin and delicate continuation of cranial arachnoid. Separate form dura by subdural space.

- It sends many delicate web-like process from its inner surface across the subarachnoid space to pia mater.
- Well developed in posterior median forming posterior median septum.

Pia mater: A delicate connective tissue highly vascular and closely applied to the cord. It covers the spinal cord and enters to line anterior median sulcus.

- It prolongs over spinal nerve roots and blends with its epineuria.
- Inferiorly projects below the apex of conus medullaris where it extends as filum terminalis.
- Filum terminalis is about 20cm long, passing through sacral hiatus of S2. It then descends back of coccyx attaching to the dorsal surface of coccyx.

The dura, arachnoid and subarachnoid space contain CSF extending down to 2nd sacral vertebral.

- Spinal cord end at L1 as cornus medullaris.
- From L1-S2, subarachnoid space contain spinal nerve roots constituting caudae equina.
- Below L2 lumber puncture is safe.

Contrast between cranial dura with spinal dura:

- Endosteal layer is lacking in the vertebral region.
- Only meningeal layer continues into the vertebral canal.
- Cranial dural mater lies closely against cranial bone with no intervening epidural space as observed around spinal cord.

Applied Anatomy: Meningitis an inflammation of meninges, a serious disease of infancy and childhood. Occurs from 3 months to 2 yrs. of age.

Caused by varieties of bacteria and viruses that invade CNS by way of nose & often follows respiratory trait or ear infection.

Arachnoid and pia maters most often infected from there infection can spread to adjacent nervous tissue.

- In bacterial meningitis the brain swells, ventricle enlarges with signs of high fever, stiff neck, drowsiness and intense headache. May progress to vomiting, loss of sensory and motor function leading to coma and death ultimately.
- Diagnosis: Lumber puncture below L2 to obtain CSF for bacteria.

Extradural and Subdural Haematomas

A haematoma is a collection of **blood**. As the cranial cavity is effectively a closed box, a haematoma can cause a rapid increase in Intra-cranial pressure resulting to death if untreated.

There are two types of haematomas involving the dura mater:

Extradural – Arterial blood collects between the skull and endosteal layer of the dura. The causative vessel is usually the middle meningeal artery, tearing as a consequence of brain trauma.

Subdural – Venous blood collects between the dura and the arachnoid mater. It results from damage to cerebral veins as they empty into the dural venous sinuses.